FrutaMIDI: capacitive USB MIDI controller

Engineering report

FrutaMIDI engineering project

Revision: `TO BE FILLED FROM EVIDENCE`

Abstract

TO BE FILLED FROM EVIDENCE. One-paragraph summary of scope, verification status and open risks, written last from the gate summaries.

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1 Introduction and scope

FrutaMIDI is a USB MIDI controller that turns touches on conductive objects (fruit, plants, water) into MIDI notes, using a Raspberry Pi Pico and an MPR121 capacitive sensor. It is built either on a breadboard or on a single-sided PCB in `hardware/`.

1.1 Scope

TO BE FILLED FROM EVIDENCE. In-scope and out-of-scope items, from management/PROJECT_DEFINITION.md.

1.2 Document conventions

- Every number carries one label: SOURCE, CALCULATION, SIMULATION, MEASUREMENT or ASSUMPTION.
- Status is reported only as PASS, PARTIAL, FAIL, BLOCKED or NOT RUN.
- Passing ERC, DRC or a compile is not hardware validation.

2 Requirements and verification status

Source: `management/REQUIREMENTS.md`. The table below is generated from a CSV export into `generated/requirements_table.tex`; do not edit it by hand.

Table 1: Requirements and verification status (generated from requirements/requirements.csv).

ID	Requirement	Verification	Status	Evidence
REQ-001	**Note On/Off per electrode** Each of the 12 electrode inputs E0–E11 (MPR121 ELE0–ELE11) shall send one MIDI Note On, on the configured channel (default 1) with the configured fixed velocity (default 100, range 1–127), when a touch is detected. It shall send a Note Off for the same note when the touch is released.	I + T [PHYSICAL]	PARTIAL	I: PASS. The chain terminal → series resistor → module electrode is 1:1 (interface_check `pcb.A2.pin2..13`, `pcb.J2.*`). T: NOT RUN.
REQ-002	**Note Off integrity** The Note Off for a pad shall carry the note number sent in its Note On, even if the scale or octave changed in between. All sounding notes shall be released before a recalibration.	T (host unit test) + T [PHYSICAL]	PARTIAL	Host tests PASS: `reports/firmware/T-040/pytest_tests_firmware.log` ("213 passed"). Hardware test NOT RUN.
REQ-003	**Second sensor (24 pads)** The firmware shall support a second MPR121 at I2C address 0x5B (`NUM_SENSORS = 2`), mapping its electrodes to pads 12–23. This is a breadboard-only option; the PCB carries one module (ADR-007 D-3).	A + T [PHYSICAL]	PARTIAL	A: PASS for latency at 400 kHz (11.50 ms $\leq$ 15 ms). A: FAIL at 100 kHz (17.98 ms) (budgets.out §b). T: NOT RUN.
REQ-004	**Polyphonic aftertouch** While a pad is touched, the device shall send Polyphonic Key Pressure (0xAn) with a value 0–127. The value shall map linearly from delta = baseline − filtered, clamped to [12, 120]. It shall be evaluated every 15 ms and sent only when it changes by at least 2.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-040, `test_aftertouch.py`). Hardware NOT RUN.

ID	Requirement	Verification	Status	Evidence
REQ-005	**Six scales** The device shall provide 6 scales, selected cyclically: Drum Rack (36–47, octave ignored, the power-on default), major pentatonic, minor pentatonic, major, natural minor and chromatic (root 48). Notes shall be clamped to 0–127.	A + T (host)	PASS	A: notes 24–99 with 12 pads and 24–127 with 24 pads, never clamped (T-009 budgets §d). Host tests PASS (T-040 `test_scales.py`).
REQ-006	**Octave potentiometer** The octave potentiometer shall shift melodic scales by −2..+2 octaves using 5 ADC zones with a hysteresis of 0.7 zone. It shall have no effect on Drum Rack, and shall be enabled by default on the PCB build.	A + T (host) + T [PHYSICAL]	PARTIAL	A: the ADC offset of 30 mV (SOURCE S-01 §4.3) is 6.5 % of the 0.462 V hysteresis (CALCULATION). Host tests PASS (`test_pot.py`). Hardware NOT RUN.
REQ-007	**Button short press** A short press of the button (released within 800 ms, sampled every 10 ms) shall select the next scale, and the LED shall blink N times for scale N.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (`test_button.py`, `test_blink.py`). Hardware NOT RUN.
REQ-008	**Button long press recalibration** A press of more than 800 ms shall release all notes and recalibrate all sensors; the LED then blinks 3 times.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS. Recalibration blocks for 16.8–22.1 ms per sensor at 400 kHz (CALCULATION); this is user-requested and allowed by REQ-016. Hardware NOT RUN.
REQ-009	**Sensor-absent behaviour** If no MPR121 answers, the device shall send no MIDI, blink the LED at 5 Hz and retry every 2 s.	T [PHYSICAL]	NOT RUN	
REQ-010	**Status LED** The LED shall give 2 slow blinks at start when a sensor is found, and shall be lit while any pad is touched. On Pico W and Pico 2 W the LED is driven through the CYW43 (WL_GPIO0).	T [PHYSICAL]	NOT RUN	
REQ-011	**USB MIDI class compliance** The device shall enumerate as a USB MIDI 1.0 class device (MIDIStreaming, bulk endpoints). It shall work without vendor drivers on Windows 10, Windows 11 and macOS 12 or later.	T [PHYSICAL]	NOT RUN	
REQ-012	**Device name** The USB product string and the MIDI port name shall be "FrutaMIDI".	I + T [PHYSICAL]	PARTIAL	I: `DEVICE_NAME = "FrutaMIDI"` is passed to `setProductDescriptor` and `setStringDescriptor`. T: NOT RUN.
REQ-013	**MIDI message set** The device shall send only Note On (0x9n, CIN 0x9), Note Off (0x8n, velocity 0, CIN 0x8) and Polyphonic Key Pressure (0xAn, CIN 0xA), on one channel n = MIDI_CHANNEL − 1. Incoming MIDI shall be read and discarded.	I	PASS	`fm_midi.h` constants match S-14 Table 4-1.

ID	Requirement	Verification	Status	Evidence
REQ-014	**Touch-to-MIDI latency** Device latency from touch to the USB-MIDI packet being available to the host shall be $\leq$ 15 ms worst case, for electrode capacitance $\leq$ 100 pF, with mitigation M1 (one burst read per sensor per aftertouch pass).	A + T [PHYSICAL]	PARTIAL	A: PASS. 1 sensor 10.30 ms, 2 sensors 11.50 ms at 400 kHz (the clock retained by ADR-008); 100 kHz would give 13.54 ms. Design limit C $\leq$ 210 pF (CALCULATION, budgets.out §b). Based on the firmware's actual transactions: `touched()` (2 B) plus burst reads of 24 B and 12 B (M1, already in FrutaMIDI.ino sha256 b3f25158). Depends on firmware T-041. T: NOT RUN (plan item P-2). Flagged for user review.
REQ-015	**Latency jitter** Latency jitter (worst minus best) shall be $\leq$ 10 ms for electrode capacitance $\leq$ 100 pF.	A + T [PHYSICAL]	PARTIAL	A: PASS. 6.30 ms at 400 kHz, 9.54 ms at 100 kHz (0.46 ms margin). T: NOT RUN.
REQ-016	**No blocking during play** During play, no firmware action other than a user-requested recalibration shall block touch processing for more than 10 ms.	I + A	PASS	I: the only `delay()` is `delay(10)` in `midiBegin()` (setup); LED feedback uses the non-blocking `fm_blink.h` (T-040). A: worst loop 1.30 ms at 400 kHz. Depends on firmware T-041, which is still being finished; re-check after it lands.
REQ-017	**USB bus power only** The device shall be powered only from USB VBUS through the Pico's micro-USB connector. No other power input shall exist.	I	PASS	Pico pins 39 and 40 are unconnected (`pcb.A1.no_other_nets`).
REQ-018	**VBUS range** The device shall operate with VBUS from 4.40 V to 5.50 V at its connector.	A + T [PHYSICAL]	PARTIAL	A: VSYS 3.95–5.25 V, inside the SOURCE S-01 range of 1.8–5.5 V: PASS. However, the 4.40 V low-power-port minimum is below the Pico's recommended VBUS of 5 V $\pm$ 10 % (4.50 V; S-01 §2.3): budgets check A3b FAIL. The diode drop is an ASSUMPTION. T: NOT RUN.
REQ-019	**VBUS current** VBUS current shall be $\leq$ 100 mA (the declared bMaxPower, S-11) for every supported Pico board.	A + T [PHYSICAL]	PARTIAL	A: worst case 55.6 / 70.6 / 60.6 / 75.6 mA for Pico / Pico W / Pico 2 / Pico 2 W. The lower bounds are SOURCE (S-01 Table 3 9.9 mA, S-16 Table 2 12.74 mA). The upper bounds are ASSUMPTIONs because neither datasheet covers a busy USB-MIDI loop at 200 MHz. T: NOT RUN (plan item P-3).
REQ-020	**3V3 rail load** The external load on the Pico 3V3(OUT) shall be $\leq$ 300 mA with at least 50 % margin.	A	PASS	10.15 mA worst case with 2 sensors and PCB pull-ups: 96.6 % margin.

ID	Requirement	Verification	Status	Evidence
REQ-021	**MPR121 supply voltage** The MPR121 VDD, fed from the Pico 3V3 through the module LDO, shall be within 2.0–3.6 V.	A	PASS	3.10–3.40 V (CALCULATION; dropout and rail tolerance are ASSUMPTIONs).
REQ-022	**Touchable voltage (SELV)** Every user-touchable conductor shall be at $\leq$ 3.6 V relative to GND and shall be supplied only from the USB SELV source.	A	PASS	Electrodes are charged from MPR121 VDD $\leq$ 3.6 V (S-04), and REQ-017 holds.
REQ-023	**Electrode ESD robustness** Each user-touched electrode terminal shall survive ±4 kV contact discharge without permanent damage. The ±4 kV target is an ASSUMPTION, following IEC 61000-4-2 level 2 (the standard was not read).	I + A + T [PHYSICAL]	PARTIAL	A: PASS at ±4 kV. With 3.3 kΩ series per electrode (R1–R12), the peak current is 1.10 A, against a 1.33 A pin limit inferred from the HBM ±2 kV rating (ASSUMPTION) (`reports/electronics/analysis.md`, ADR-008). My own check gives the same 1.10 A (CALCULATION). At 8 kV the peak is 2.20 A (1.65× over): PARTIAL, accepted as documented risk R-006. The resistor pulse rating is not checked. Sensing impact: sensitivity factor 0.904 (EE); τ/CDT $\leq$ 0.10; I·R offset 0.208 V (CALCULATION). T: NOT RUN [PHYSICAL]; it needs an ESD generator.
REQ-024	**I2C hand-wire fault tolerance** A misplaced SDA or SCL hand wire shall not connect an I2C line to a supply hole.	I + A	FAIL	I (interface_check, current PCB): the SDA/SCL hand wires from J1 (1 = SCL, 2 = SDA) can reach S1/S7 (+3V3 when JP1 is bridged) and S2/S6 (module 3.3 V output). An I2C low then drives the RP2040 pad against a supply; the current is limited only by the pad driver (ASSUMPTION). T-052 did not change this. Mitigation option: 220 Ω series resistors in SDA/SCL, limiting the fault current to $\leq$ 15 mA (CALCULATION). Open decision D-9.
REQ-025	**Both module revisions** One PCB shall accept both Adafruit 1982 revisions. JP1 bridges ORIG (+3V3 → S1 = VIN) or QT (+3V3 → S7 = VCC). The SDA/SCL wires from J1 go to S5/S6 (original) or S3/S4 (STEMMA QT).	I + T [PHYSICAL]	PARTIAL	I: PASS (`pcb.JP1_selector`, `pcb.JP1_targets`, the S-row labels, `pcb.J1`). T: NOT RUN.

ID	Requirement	Verification	Status	Evidence
REQ-026	**I2C rise time (PCB)** The PCB-build I2C bus shall meet the rise time for its clock: $\leq$ 300 ns at 400 kHz ($\leq$ 1000 ns at 100 kHz), with the module at 0x5A.	A + T [PHYSICAL]	PARTIAL	A: PASS. R13/R14 4.7 kΩ: electronics gives 103 ns at 38 pF and 135 ns at 50 pF (`reports/electronics/analysis.md`). My check, which also includes the module 10 kΩ and a 50 kΩ internal pull-up, gives 153 ns at 60 pF, with a C_b limit of 117.8 pF at 400 kHz; sink current 1.43 mA $\leq$ 3 mA (CALCULATION). The clock stays at 400 kHz (ADR-008, D-5 closed). T: NOT RUN (plan item P-4).
REQ-027	**Single-sided PCB** The PCB shall be single-sided: copper on B.Cu only, through-hole parts only, no vias, no copper pour, and no wire bridges apart from the 2 I2C configuration wires and the JP1 solder bridge.	I	PASS	pcb_rules_check: all segments on B.Cu; 0 vias, 0 zones, 0 SMD pads.
REQ-028	**Board outline** The board outline shall be 100 × 70 mm.	I	PASS	Outline 100.000 × 70.000 mm. Separate finding: the PCB silkscreen still reads 'FrutaMIDI v1.0' where ADR-008 requires v1.1 (interface_check `pcb.label_v1.1` FAIL).
REQ-029	**Home-etch rules** Home-etch rules: signal tracks $\geq$ 1.0 mm ($\geq$ 0.8 mm for short stubs), power tracks $\geq$ 1.2 mm, clearance $\geq$ 0.6 mm (DRC hard minimum 0.5 mm), annular ring $\geq$ 0.4 mm, drills $\geq$ 0.8 mm.	I + A (DRC)	PASS	pcb_rules_check `SUMMARY: 14 checks, 14 PASS, 0 FAIL`; minimum annular ring 0.40 mm. Clearance relies on the electronics DRC (ERC 0, DRC 0, unconnected 0, parity 0; manager re-run, `reports/electronics/analysis.md`), which systems did not re-run.
REQ-030	**Screw-terminal map** The PCB shall provide 14 screw terminals at 5.08 mm pitch: GND, E0..E11, GND. Terminal k (k = 2..13) shall reach MPR121 electrode k − 2 through its series resistor.	I	PASS	interface_check J2.T1..T14 and A2.pin2..13.
REQ-031	**Pin map consistency** The firmware PCB pin map (I2C1 SDA GP26 / SCL GP27, button GP22, pot GP28/ADC2) shall match the PCB nets. The breadboard map (I2C0 GP4/GP5, button GP15, pot GP26) shall match README.md.	I	PASS	interface_check: 0 pin mismatches.
REQ-032	**Core BOM cost** The core BOM (assembled board, excluding tools, consumables and USB cable) shall cost $\leq$ 30 EUR.	A	FAIL	20.38–30.30 EUR (CALCULATION). Nearly all prices are ASSUMPTIONs; the upper bound exceeds the target by 0.30 EUR. Quoted prices from one supplier would close this.

ID	Requirement	Verification	Status	Evidence
REQ-033	**Firmware build matrix** The firmware shall build without errors for Pico, Pico W, Pico 2 and Pico 2 W, each for the breadboard and PCB variants (FQBN `rp2040:rp2040:<board>:usbstack=tinyusb`, plus `-DFRUTAMIDI_PCB` for PCB), and for `arduino:avr:leonardo` (optional).	T (compile)	PASS	`reports/firmware/T-041/build/*.log`: 9 of 9 logs contain "Sketch uses . . . ". A clean compile is not a hardware validation.
REQ-034	**Browser sampler** The browser sampler `docs/index.html` shall work in Chrome and Edge (Web MIDI) on Windows and macOS. It shall map its 12 pads to notes 36–47 by default, treat Note On with velocity 0 as Note Off, send unmapped notes to the synthesiser, apply aftertouch (0xA0) to the filter, and offer a keyboard fallback and dropped-file samples.	T	PARTIAL	Automated browser tests with a stubbed Web MIDI PASS (T-020, T-021 handoffs). A real browser, a real device and audible output are NOT RUN.
REQ-035	**English language** All project files, code, comments, reports, generated PCB text, fabrication file names and user-visible strings shall be in English.	I	PARTIAL	Now English: firmware (`.ino`, `fm_*.h`), gen_*.py, PCB silkscreen ("SCALE", "OCTAVE", "ORIG", "QT", "copper: bottom side"), `hardware/fabrication/` file names, docs/index.html (0 Portuguese matches in a scan). Remaining: README.md and hardware/README.md link to the old Portuguese paths `fabrico/...` (broken links, `doc.links.*` FAIL) and quote the silkscreen as "ESCALA"/"OITAVA".
REQ-036	**User and assembly manual** The user and assembly manual shall match the current PCB. It shall cover both module revisions, the JP1 selector and data wires, flashing, first start, a valid pre-power continuity checklist and troubleshooting.	I	FAIL	As of this snapshot, hardware/README.md still describes the J1 3V3 wire (no JP1). Its continuity test expects a beep from terminal to module, which now passes through 3.3 kΩ. Its links are broken (interface_check `doc.hw_readme.*`, 7 FAIL). This may be work in progress by the documentation engineer.
REQ-037	**Sensing range (physical)** [PHYSICAL] Touch detection shall work on objects from a bare pad (about 20 pF) to large fruit or a glass of water (up to about 500 pF, ASSUMPTION), with leads of 20–50 cm, after recalibration.	T [PHYSICAL]	NOT RUN	A: the MPR121 can charge up to about 1203 pF (CALCULATION). The latency targets hold up to 210 pF at 400 kHz.
REQ-038	**False triggers and crosstalk (physical)** [PHYSICAL] There shall be no false Note On in 10 min idle with all 12 electrodes connected to objects, and no crosstalk when touching a neighbouring electrode.	T [PHYSICAL]	NOT RUN	
REQ-039	**JP1 cannot short 3V3** No position of the JP1 3V3 selector (ORIG, QT, open) shall connect +3V3 to GND.	I	PASS	S1 and S7 are never GND in either revision (`pcb.JP1_no_gnd_reachable`).

ID	Requirement	Verification	Status	Evidence
REQ-040	**JP1 wrong-position tolerance** A wrong JP1 position (ORIG on a STEMMA QT module, or QT on an original module), or both sides bridged, shall not apply a voltage outside the MPR121 absolute maximum ratings (inputs $\leq$ VDD + 0.3 V, S-04 Table 3), and shall not short +3V3 through a module output.	A	FAIL	The ORIG bridge on a QT module puts +3V3 on ADDR while the module is unpowered. The QT bridge on an original module puts +3V3 on IRQ while the module is unpowered. Both exceed VDD + 0.3 V, back-powering the module through its input clamp. Bridging both sides on an original module ties +3V3 to the open-drain IRQ output, which shorts through the MPR121 when IRQ asserts. Expected outcome: sensor not found; the damage risk is low but unquantified (ASSUMPTION; S-04 latch-up rating ±100 mA). Open decision D-10.
REQ-041	**I2C rise time (breadboard)** The breadboard-build I2C bus (no PCB pull-ups) shall meet the rise time for its clock.	A + T [PHYSICAL]	FAIL	A: on the breadboard (no R13/R14), R_eff 8.33 kΩ gives t_r 424 ns at 60 pF, which fails the 300 ns limit at 400 kHz (C_b limit 42.5 pF). ADR-008 keeps 400 kHz, so the breadboard build needs the 4.7 kΩ pull-ups documented in the README, or a measurement (P-4). Open decision D-11.

Status counts (41 requirements): PASS 14, PARTIAL 17, FAIL 5, NOT RUN 5.

3 Architecture and decisions

3.1 System overview

TO BE FILLED FROM EVIDENCE. Block diagram and description: electrodes, MPR121, Pico, USB MIDI, host sampler/DAW. Figure from reports/figures/.

3.2 Decisions

TO BE FILLED FROM EVIDENCE. Summary of each ADR in management/DECISIONS.md: context, decision, consequences, status.

4 Interfaces

Source: `management/INTERFACES.json`.

TO BE FILLED FROM EVIDENCE. Electrical interfaces (pins, voltages, I^2C addresses), USB MIDI messages, and host interfaces. Each value labelled.

5 Electronics analysis

5.1 Schematic and PCB

TO BE FILLED FROM EVIDENCE. Design summary, ERC/DRC results with report paths and revision. ERC/DRC passing is not hardware validation.

5.2 Analyses

TO BE FILLED FROM EVIDENCE. Power budget, sensing range, ESD and current limiting on touchable electrodes (T-050), with assumptions, sensitivity and, where applicable, convergence.

5.3 Safety statement

TO BE FILLED FROM EVIDENCE. Final wording of the safety statement (ADR-003), based on the analysis above.

6 Firmware design and tests

Source: `firmware/FrutaMIDI/`.

6.1 Design

TO BE FILLED FROM EVIDENCE. Module structure, main loop, touch/release handling, scales, octave, aftertouch, recalibration.

6.2 Build and tests

TO BE FILLED FROM EVIDENCE. Compile results per board (tool version, result line), host unit tests (command, pass count), with log paths and revision.

7 Host software and tests

Source: `docs/index.html` (browser sampler).

TO BE FILLED FROM EVIDENCE. Sampler design, MIDI handling, tests performed (command, result line), browser compatibility evidence.

8 Independent review findings

TO BE FILLED FROM EVIDENCE. Findings from the independent reviewer: ID, severity, description, disposition (fixed / accepted / open), evidence path.

9 Limitations

TO BE FILLED FROM EVIDENCE. Known limitations, untraced claims, assumptions not yet checked and open risks from management/RISKS.md.

10 Physical validation plan

No part of the hardware has been measured. Each row below defines a physical test still to run. Pass criteria come from the requirements, not from this report.

What to measure	How (instrument, setup)	Pass criterion	Status
TO BE FILLED FROM EVIDENCE			NOT RUN

11 Reproducibility

11.1 Toolchain

TO BE FILLED FROM EVIDENCE. Tool versions from management/TOOLCHAIN.md (KiCad, arduino-cli and cores, Python, Node, TinyTeX and installed packages).

11.2 How to rebuild

- Hardware: `sh hardware/scripts/build_all.sh` (regenerates, runs ERC/DRC, exports fabrication files).
- Report: MCP `eng-tools latex_build` with `reports/report/frutamidi_report.tex`.

TO BE FILLED FROM EVIDENCE. Firmware compile and test commands, released file checksums (releases/V1_BASELINE/).

A Sources

Source list: `management/SOURCES.md`.

TO BE FILLED FROM EVIDENCE. Datasheets, standards and external references cited in this report, with identifier and access date.